

WEST PENOBSCOT BAY BUOY 44033, ME, USA

WMO: 997180

Lat: 44.055N

Lon: 68.996W

Elev: 0

StdP: 14.70

Time Zone: -5.00 (NAE)

Period: 05-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	7.9	12.1	N/A	N/A	N/A	N/A	N/A	N/A	36.9	23.4	33.9	26.4	17.1	300	0.666

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.7	69.9	N/A	67.8	N/A	66.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	8.2	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 31.4	26.5	24.0	DB	1.8	76.0	4.3	2.9	-1.3	78.1	-3.8	79.8	-6.2	81.5	-9.3	83.6	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.0	27.9	28.0	33.0	42.1	49.4	56.4	61.8	62.5	59.3	52.0	44.2	34.6
	DBStd	13.54	9.22	7.35	6.39	4.07	3.97	3.70	3.47	2.34	3.51	4.81	6.53	8.18
	HDD50	2828	686	615	526	236	59	2	0	0	0	36	191	477
	HDD65	6942	1151	1035	991	686	483	258	110	84	174	403	625	942
	CDD50	1381	0	0	0	1	41	194	366	387	278	99	15	0
	CDD65	18	0	0	0	0	0	0	10	5	3	0	0	0
	CDH74	7	0	0	0	0	0	0	6	1	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	11.8	15.7	14.7	14.3	11.7	9.9	8.6	7.3	7.5	9.7	13.1	14.2	15.5
Precipitation	PrecAvg	50.7	4.1	3.5	4.4	4.6	4.1	4.1	3.4	3.0	4.4	5.3	5.3	4.9
	PrecMax	73.1	7.6	8.0	8.4	9.9	9.9	9.8	7.3	7.0	10.0	15.7	9.3	8.8
	PrecMin	30.2	0.5	1.2	1.1	0.4	0.7	2.0	1.4	0.8	1.3	1.3	2.6	1.9
	PrecStd	9.4	1.8	1.4	1.9	2.4	2.0	2.1	1.7	1.7	2.5	3.1	1.5	1.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.5	43.9	47.3	54.8	66.1	68.7	73.8	71.5	70.9	64.1	57.1	52.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	45.7	41.3	43.8	50.8	59.7	65.2	70.6	68.6	67.2	60.5	55.2	50.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	43.5	39.7	41.9	48.6	56.8	63.0	68.2	67.0	65.1	59.2	53.8	48.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	41.3	38.1	40.4	47.2	54.6	61.3	66.4	65.8	63.6	58.0	52.8	46.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
Mean Daily Temperature Range	5% DB	MDBR	11.0	9.8	8.5	6.4	6.3	6.2	6.7	5.7	5.8	6.8	7.5	9.7
		MCDBR	11.0	9.6	8.8	9.0	10.1	9.8	9.8	7.5	7.1	6.8	7.1	10.9
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.299	0.307	0.329	0.355	0.381	0.423	0.442	0.411	0.368	0.339	0.320	0.299	
	taud	2.473	2.464	2.464	2.425	2.395	2.293	2.262	2.337	2.439	2.518	2.535	2.516	
	Ebn at Noon	264	282	289	289	282	270	263	267	272	267	254	253	
	Edh at Noon	23	27	31	35	37	41	42	37	31	25	21	20	
All-Sky Solar Radiation	RadAvg	503	787	1169	1464	1663	1792	1877	1689	1361	874	542	399	
	RadStd	39	63	74	155	199	151	135	89	81	80	49	43	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (33 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+108	N/A

Nomenclature: See separate page